

Hackathon Prompt: Quantum Coin Flipper

Background and motivation:

Imagine you are designing a game that needs a perfectly fair coin flip.

A classical computer can simulate a coin flip using a pseudo-random number generator. But can we create a coin flip using the fundamental randomness of quantum mechanics?

In this challenge, you will build a **quantum coin**.

Getting started:

Your task is to create a one-qubit quantum circuit that behaves like a fair coin.

Your circuit should produce:

- 0 approximately 50% of the time
- 1 approximately 50% of the time

Use a **Hadamard gate** to create the required quantum state.

Then measure your qubit.

Your Mission:

Create a program that prints:

- Quantum Coin: HEADS

or

- Quantum Coin: TAILS

based on the measurement result.



Experiment:

Run your circuit:

- 10 times
- 100 times
- 1,000 times
- 10,000 times

Record the percentage of heads and tails.

Deeper Questions:

Why doesn't 1,000 shots necessarily produce exactly 500 heads and 500 tails?

What happens if you apply another quantum gate before measurement?

Can you intentionally create a biased quantum coin?

Stretch Challenge:

Create a game where two players call heads or tails before the quantum coin is flipped.

Keep track of the scores and determine the winner.

